

Your grade: **100%**

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Next item →

1. Which of the following are necessary steps in developing a deep learning model using Keras?

1 / 1 point

- ☐ Train the model on the test set
- ☐ Deploy the model using `deploy()`
- ☒ Initialize the optimizer

✔ **Correct**
Yes, initializing the optimizer is a crucial step in setting up the model.

- ☒ Import necessary libraries

✔ **Correct**
Correct! Importing necessary libraries is the first step.

- ☒ Evaluate the model using `evaluate()`

✔ **Correct**
Correct! Evaluating the model is important to understand its performance.

2. What is the primary goal when analyzing and visualizing a dataset for predicting red wine quality?

1 / 1 point

- ☐ To ensure the dataset is perfectly balanced
- ☒ To identify patterns and correlations within the data
- ☐ To apply all possible feature scaling techniques
- ☐ To remove all outliers from the dataset

✔ **Correct**
Correct! Identifying patterns and correlations helps in understanding the importance of different features for prediction.

3. What are the benefits of visualizing a dataset before building a model?

1 / 1 point

- ☐ Automatically selects the best model for the data
- ☒ Helps identify any missing values

✔ **Correct**
Correct! Visualization can help in identifying missing values that need to be addressed.

- ☐ Directly improves model accuracy
- ☒ Allows for better understanding of feature distributions

✔ **Correct**
Correct! Visualization aids in understanding how the features are distributed.

- ☒ Detects outliers and anomalies

✔ **Correct**
Correct! Visualization can help in spotting outliers and anomalies that may affect model performance.

4. When would you use a KDE plot instead of a box plot?

1 / 1 point

- ☒ To estimate the probability density function of a continuous variable
- ☐ To show the relationship between two categorical variables
- ☐ To display the frequency of categorical data
- ☐ To identify outliers in the data

✔ **Correct**
Correct! A KDE plot is used to estimate the probability density function of a continuous variable, providing a smooth curve.

5. What information does the `'describe()'` method provide in exploratory data analysis?

1 / 1 point

- ☐ Data types of each column
- ☐ The first five rows of the DataFrame
- ☒ Summary statistics like mean, median, and standard deviation
- ☐ The shape of the DataFrame

✔ **Correct**
Correct! The `'describe()'` method gives a summary of statistics for numerical columns, including mean, median, standard deviation, and more.